

Math 231B HW 3

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Problem 1

Etingof Lecture Note Exercise 31.15:

- (i) Show that for even $\dim V$, the representations $(\xi^*M)_0, (\xi^*M)_1$ are isomorphic to S_+, S_- respectively.
- (ii) Show that for odd $\dim V$, the representations ξ^*M_+ and ξ^*M_- are both isomorphic to S .

Hint: Find the highest weight vector for each of these representations and compute the weight of this vector. Then compare dimensions.

Solution:

- (i) For even dimension case, recall that V has basis $a_1, \dots, a_n, b_1, \dots, b_n$, while $M = \bigwedge(a_1, \dots, a_n)$. Now, consider the following elements in $\bigwedge^2 V \cong \mathfrak{so}(V)$:

$$\forall i \in \{1, \dots, n\}, \quad h_i := a_i \wedge b_i \quad (1.1)$$

Which, inside the Clifford Algebra $\text{Cl}(V)$, the corresponding action is $\xi(a_i \wedge b_i) = a_i b_i - \frac{1}{2}(a_i, b_i) = a_i b_i - \frac{1}{2}$.

Top Wedge Case:

If consider the action on the top wedge $(a_1 \dots a_n) \in M$, then one has:

$$\begin{aligned} h_i * (a_1 \dots a_n) &= \left(a_i b_i - \frac{1}{2} \right) * (a_1 \dots a_n) \\ &= a_i * (-1)^{i-1} (a_1 \dots \hat{a}_i \dots a_n) - \frac{1}{2} (a_1 \dots a_n) \\ &= (a_1 \dots a_n) - \frac{1}{2} (a_1 \dots a_n) \\ &= \frac{1}{2} (a_1 \dots a_n) \end{aligned} \quad (1.2)$$

This shows that with respect to all h_i 's, $(a_1 \dots a_n)$ has been acted by multiplying by $\frac{1}{2}$. So, its weight is the fundamental weight $w_{n-1} = (\frac{1}{2}, \dots, \frac{1}{2})$ for $\mathfrak{so}(2n)$.

Also, notice that since any $b_i b_j = \xi(b_i \wedge b_j)$ (for $i \neq j$), then no matter if $a_1 \dots a_n \in (\xi^*M)_0$ or $(\xi^*M)_1$ (corresponds to the case where n is even / odd respectively), it generates everything, as if $a_{i_1} \dots a_{i_k}$ is a wedge with $n - k$ being even (i.e. they have the same parity), if $a_{j_1}, \dots, a_{j_l}$ are distinct variables that didn't show up in $a_{i_1} \dots a_{i_k}$ (say $l = n - k$), then notice the following:

$$\prod_{i=1}^{\frac{l}{2} = \frac{n-k}{2}} b_{j_{2i}} b_{j_{2i+1}} * (a_1 \dots a_k) = (-1)^L (a_{i_1} \dots a_{i_l}) \quad (1.3)$$

since $b_{j_1} \dots b_{j_l}$ cancels all appearance of $a_{j_1}, \dots, a_{j_l}$ in $(a_1 \dots a_n)$, and left with $a_{i_1} \dots a_{i_k}$ (up to some scalars). So, $a_1 \dots a_n$ is indeed a highest weight vector of $(\xi^* M)_0$ or $(\xi^* M)_1$ with fundamental weight $w_{n-1} = (\frac{1}{2}, \dots, \frac{1}{2})$ (dependent on the even / oddness of n), and generates everything).

Second Top Wedge Case:

If consider the second top wedge $a_1 \dots a_{n-1}$ (if $a_1 \dots a_n$ belongs to one of $(\xi^* M)_0, (\xi^* M)_1$, then $a_1 \dots a_{n-1}$ must be in the other), notice that each h_i acts as below:

$$h_i * (a_1 \dots a_{n-1}) = \left(a_i b_i - \frac{1}{2} \right) * (a_1 \dots a_{n-1}) = \begin{cases} \frac{1}{2} (a_1 \dots a_{n-1}) & i < n \\ -\frac{1}{2} (a_1 \dots a_{n-1}) & i = n \end{cases} \quad (1.4)$$

(Note: the reason is because if $i < n$, then a_i shows up in $a_1 \dots a_{n-1}$, so $a_i b_i$ acts as 1; on the other hand, if $i = n$, $a_n b_n$ acts on $a_1 \dots a_{n-1}$ as 0, since it doesn't have a_n term, so the $-\frac{1}{2}$ in h_i takes over).

Hence, $(a_1 \dots a_{n-1})$ in fact corresponds to the weight $w_n = (\frac{1}{2}, \dots, \frac{1}{2}, -\frac{1}{2})$ for $\mathfrak{so}(2n)$.

Finally, this wedge also generates everything with the same degree parity: if take any other $a_1 \dots \hat{a}_i \dots a_n$ (where $i \neq n$), then one has $(a_n b_i) * (a_1 \dots a_{n-1}) = (-1)^k (a_1 \dots \hat{a}_i \dots a_n)$ (by cancelling a_i , and add a_n , up to a scalar).

Also, similar to the top wedge case, if any other wedge is with even degree difference with $a_1 \dots a_{n-1}$, first move to the suitable $a_1 \dots \hat{a}_i \dots a_n$ (such that the desired wedge also misses a_i), then lower using the combinations of $b_j b_k$'s.

This shows that if $(a_1 \dots a_{n-1})$ lies in $(\xi^* M)_0$ or $(\xi^* M)_1$ (again, based on the odd/evenness of n), it's a highest weight of fundamental weight w_n that generates everything.

Finally, for the above two cases, notice that $\dim(\xi^* M)_0 = \sum_{0 \leq 2k \leq n} \binom{n}{2k}$ (sum of all distinct even degree wedges), and $\dim(\xi^* M)_1 = \sum_{0 \leq 2k+1 \leq n} \binom{n}{2k+1}$ (sum of all choices of distinct odd degree wedges). But, if consider the following binomial expansion:

$$(x+1)^n = \sum_{k=0}^n \binom{n}{k} x^k \quad (1.5)$$

Plugin $x = -1$, one yields the following:

$$0 = \sum_{k=1}^n \binom{n}{k} (-1)^k = \sum_{0 \leq 2k \leq n} \binom{n}{2k} - \sum_{0 \leq 2k+1 \leq n} \binom{n}{2k+1} \quad (1.6)$$

This equation shows that $\dim(\xi^* M)_0 = \dim(\xi^* M)_1$. And, with the two direct sums to $M = \bigwedge(a_1 \dots a_n)$ (which has dimension 2^n), each of them must have dimension 2^{n-1} , matching up with the dimension of S_+, S_- .

Since the weight of the highest weight representations match up, together with the dimension matches up, one concludes that $(\xi^* M)_0, (\xi^* M)_1$ are isomorphic to S_+, S_- (depending on the parity of n).

- (ii) For odd dimension case, given V with basis $a_1, \dots, a_n, b_1, \dots, b_n, z$ that satisfies $(a_i, a_j) = (b_i, b_j) = (a_i, z) = (b_i, z) = 0$, $(a_i, b_j) = \delta_{ij}$, and $(z, z) = 2$. We'll define the following elements in $\mathfrak{so}(V) \cong \bigwedge^2 V$:

$$\forall i \in \{1, \dots, n\}, \quad h_i := a_i \wedge b_i \quad (1.7)$$

Now, for each case, consider the action of h_i on the top wedge $(a_1 \dots a_n) \in M_{\pm}$ respectively:

$$h_i * (a_1 \dots a_n) = \left(a_i b_i - \frac{1}{2}\right) * (a_1 \dots a_n) = (a_1 \dots a_n) - \frac{1}{2}(a_1 \dots a_n) = \frac{1}{2}(a_1 \dots a_n) \quad (1.8)$$

So, each h_i acts as a scalar of $\frac{1}{2}$ on the top wedge $(a_1 \dots a_n) \in M_{\pm}$, showing it's a weight vector corresponding to weight $w_n = (\frac{1}{2}, \dots, \frac{1}{2})$ for $\mathfrak{so}(2n+1)$. Which, notice that the element $b_i \wedge z \in \bigwedge^2 V$ satisfies:

$$\xi(b_i \wedge z) * (a_1 \dots a_n) = b_i z * (a_1 \dots a_n) = b_i * (-1)^n (a_1 \dots a_n) = (-1)^{n+i-1} (a_1 \dots \hat{a}_i \dots a_n) \quad (1.9)$$

Which, for any $a_{i_1} \dots a_{i_k} \in M$, if it's missing $a_{j_1}, \dots, a_{j_l}$ in the wedge, then the above formula implies $(\prod_{k=1}^l b_{j_k} z) * (a_1 \dots a_n) = (-1)^L (a_{i_1} \dots a_{i_k})$ (since it omits the $a_{j_1}, \dots, a_{j_l}$ up to a sign).

Hence, $(a_1 \dots a_n)$ in fact generates the whole $M_{\pm}$ (since the z -action here doesn't matter, only up to ± 1 on each pure wedge). Since each has dimension 2^n , and it's a highest weight module for fundamental weight $w_n = (\frac{1}{2}, \dots, \frac{1}{2})$, then both $M_{\pm}$ must agree with the spin representation S .

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Problem 2

Etingof Lecture Notes Exercise 35.8 (i), (ii):

Let $G = \mathrm{GL}_n(\mathbb{C})$. A **regular algebraic function** on G is a polynomial of X_{ij} and $\det(X)^{-1}$ for $X \in G$. Denote by $\mathcal{O}(G)$ the algebra of regular algebraic functions on G .

- (i) Show that $G \times G$ acts on $\mathcal{O}(G)$ by left and right multiplication.
- (ii) (Algebraic Peter-Weyl theorem) Show that as a $G \times G$ -module, we have

$$\mathcal{O}(G) = \bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^* \quad (2.1)$$

Hint: Compute $\mathrm{Hom}_G(V, \mathcal{O}(G))$ where G acts on $\mathcal{O}(G)$ by right translations. For this, interpret elements of this space as equivariant functions $G \rightarrow V^*$ and show that such functions are automatically regular algebraic.

Solution:

- (i) Given any pair $(A, B) \in G \times G$, and algebraic function $f(X) \in \mathcal{O}(G)$ (which f is a polynomial with indeterminates X_{ij} and $\det(X)^{-1}$, for any $X \in G = \mathrm{GL}_n(\mathbb{C})$), define the action $((A, B) * f)(X) := f(A^{-1}XB)$.

First, to show it's well-defined, notice that both A^{-1}, B are $n \times n$ matrices, hence for any $X \in \mathrm{GL}_n(\mathbb{C})$, one has the entries $(A^{-1}XB)_{il} = \sum_{j=1}^n \sum_{k=1}^n (A^{-1})_{ij} X_{jk} B_{kl}$, which the entries of $A^{-1}XB$ are all linear combinations of the entries of X ; on the other hand, one has $\det(A^{-1}XB)^{-1} = (\det(A) \cdot \det(B)^{-1}) \det(X)^{-1}$, which is a multiple of $\det(X)^{-1}$. Hence, with f being a polynomial in entries of the matrix and its determinant's inverse, one has $f(A^{-1}XB)$ still be a polynomial in entries of X and $\det(X)^{-1}$ (since inputs of f now becomes linear combinations of X_{ij} 's, and multiples of $\det(X)^{-1}$, which is still a polynomial in terms of X_{ij} 's and $\det(X)^{-1}$).

Then, to show it's an action, for any $(A, B), (A', B') \in G \times G$, one has the following:

$$\begin{aligned} ((A, B) * ((A', B') * f))(X) &= ((A', B') * f)(A^{-1}XB) = f(A'^{-1}(A^{-1}XB)B') \\ &= f((AA')^{-1}X(BB')) = ((AA', BB') * f)(X) \\ &= (((A, B) \cdot (A', B')) * f)(X) \end{aligned} \quad (2.2)$$

This shows that $(A, B) * ((A', B') * f) = ((A, B) \cdot (A', B')) * f$, which it's in fact a (left) group action.

- (ii) Notice that G is a quasi-affine variety, and V^* is an affine variety, for any V that's a finite-dimensional irreducible representation of V .

First, recall that $\mathrm{Hom}_G(V, \mathcal{O}(G))$ are all linear maps preserving the G -representations. As a result, if look at the symmetric tensor algebra SV , notice that it is equivalent to $\mathcal{O}(V^*)$ (all algebraic functions on V^*): If viewing V^* as some traditional $\mathbb{C}$ -vector space, like $\mathbb{C}^m$, choose a basis $x_1, \dots, x_m \in V$ and the corresponding dual basis $\varphi_1, \dots, \varphi_m \in V^*$, then each linear functional $\varphi \in V^*$ has a unique representative $\varphi = a_1\varphi_1 + \dots + a_m\varphi_m$, which $x_i : V^* \rightarrow \mathbb{C}$ by $x_i(\varphi) := \varphi(x_i) = a_i\varphi_i(x_i) = a_i$ is the „degree 1 polynomial“ associated to it.

As a result, $SV = \mathbb{C}[x_1, \dots, x_m]$ is precisely the coordinate ring of $V^* \cong \mathbb{C}^m$ (or $\mathbb{C}[x_1, \dots, x_m] \cong \mathcal{O}(V^*)$). Hence, with each morphism in $\mathrm{Hom}_G(V, \mathcal{O}(G))$ indicates a specific way of mapping $x_1, \dots, x_m$ into $\mathcal{O}(G)$ (which is in fact a $\mathbb{C}$ -algebra, as it's a polynomial algebra over $\mathbb{C}$ on the quasi-affine variety G), hence they each correspond to a unique algebra homomorphism $\mathbb{C}[x_1, \dots, x_m] = \mathcal{O}(V^*) \rightarrow \mathcal{O}(G)$, that preserves the G -action on the tensor algebra $S = \mathbb{C}[x_1, \dots, x_m]$ and on $\mathcal{O}(G)$ (since the morphism mapping $x_1, \dots, x_m$

to $\mathcal{O}(G)$ preserves each generator's G -action). So, there is a one-to-one correspondance $\text{Hom}_G(V, \mathcal{O}(G)) \leftrightarrow \text{Hom}_{\mathbb{C}\text{-Alg}}^{G\text{-Set}}(\mathcal{O}(V^*), \mathcal{O}(G))$ (indicating all $\mathbb{C}$ -algebra homomorphism between $\mathcal{O}(V^*)$ and $\mathcal{O}(G)$, that is also a G -equivariant map).

Now, notice that this has a one-to-one correspondance to morphisms $\text{Hom}_{G\text{-Set}}(G, V^*)$, where this is also a subset of the morphisms between affine varieties (since all the morphism $f : G \rightarrow V^*$ that satisfies $f^* \in \text{Hom}_{\mathbb{C}\text{-Alg}}^{G\text{-Set}}(\mathcal{O}(V^*), \mathcal{O}(G))$ must satisfy $f^*(g)(XA) = A * f^*(g)(X) = f^*(g \circ \rho_V(A)^{-1})(X)$ for all $g \in \mathcal{O}(V^*)$, and $A, X \in G$ for the G -action property, so one has $g \circ f(XA) = (g \circ \rho_V(A)^{-1})(f(X)) = (A * g)(f(X))$, in particular showing that f must be a G -equivariant map). So, this establishes a vijection between $\text{Hom}_G(V, \mathcal{O}(G)) \leftrightarrow \text{Hom}_{G\text{-Set}}(G, V^*)$. However, by the irreducibility, one simply has $\text{Hom}_{G\text{-Set}}(G, V^*) \cong V^*$, so one can generate an isomorphism $V \otimes V^* \cong V \otimes \text{Hom}_{G\text{-Set}}(G, V^*) \cong V \otimes \text{Hom}_G(V, \mathcal{O}(G))$.

As a result, there is a natura evaluation $\bigoplus_{V \in \text{Irrep}(G)} V \otimes V^* \cong \bigoplus_{V \in \text{Irrep}(G)} V \otimes \text{Hom}_G(V, \mathcal{O}(G)) \hookrightarrow \mathcal{O}(G)$ via the map $v \otimes f \mapsto f(v)$, for any $v \otimes f \in V \otimes \text{Hom}_G(V, \mathcal{O}(G))$ (where the injection is by the assumption of each component's irreducibility).

Yet, this map is also a surjection, because for any polynomial $f \in \mathcal{O}(G)$, which has a bounded degree in both $\det(X)^{-1}$, and the total degree of X_{ij} 's. Then, if one collects $\text{span}\{g \cdot f \in \mathcal{O}(G) \mid g \in G\}$, notice that every $g \cdot f$ must have the same precise degree in $\det(X)^{-1}$ and the total degree of X_{ij} 's (since g scales $\det(X)^{-1}$ linearly, while mapping linear functions of X_{ij} 's into each of the input of f , which both preserve the degrees). Then, in particular $Z = \text{span}\{g \cdot f \in \mathcal{O}(G) \mid g \in G\}$ is a finite-dimensional subspace of $\mathcal{O}(G)$ due to the bounded degree of all polynomials in there. And, because it collects all G -orbits of f (and their scalings), it's actually a finite-dimensional G -representation over $\mathbb{C}$.

Now, using the complete reducibility of the representation of $G = \text{GL}_n(\mathbb{C})$, then $Z \cong \bigoplus_i V_i$ for finitely many irreducible V_i 's that are G -representations. As a result, take the subspace $\bigoplus_i V_i \otimes V_i^*$ in $\bigoplus_{V \in \text{Irrep}(G)} V \otimes V^*$, since its evaluation must contain f (because f is in the direct sum of V_i 's), then f must appear in the image of the map $\bigoplus_{V \in \text{Irrep}(G)} V \otimes V^* \cong \bigoplus_{V \in \text{Irrep}(G)} V \otimes \text{Hom}_G(V, \mathcal{O}(G)) \hookrightarrow \mathcal{O}(G)$. This shows the surjection.

As a result, $\mathcal{O}(G) \cong \bigoplus_{V \in \text{Irrep}(G)} V \otimes V^*$.

Other approach to surjection:

If consider the 1-dimensional right action of $G \rightarrow \mathbb{C}^\times$ by $A \mapsto \det(A)^{-1}$, then its action on the 1-dimensional dual space $\mathbb{C}^*$ can be given by $A \cdot \lambda(_) = \lambda(_ \cdot \det(A)^{-1})$. Which, if consider the evaluation of $\mathbb{C} \otimes \mathbb{C}^*$ in $\mathcal{O}(G)$, one has the following (by choosing $a \in \mathbb{C}$ and $\lambda \in \mathbb{C}^*$ with $\lambda(a) = 1$):

$$\begin{aligned} a \otimes \lambda &\mapsto \lambda(a \cdot \det(_)^{-1}) \in \mathcal{O}(G) \\ \forall X \in G, \quad \lambda(a \cdot \det(X)^{-1}) &= \det(X)^{-1} \lambda(a) = \det(X)^{-1} \end{aligned} \quad (2.3)$$

So, $\det(X)^{-1}$ is in the image of the embedding.

On the other hand, if consider $V := \mathbb{C}^n$ as the irreducible representation of $G = \text{GL}_n(\mathbb{C})$, then with $e_1, \dots, e_n \in V$ be the standard basis, $\varphi_1, \dots, \varphi_n \in V^*$ be the corresponding dual basis, notice that the following is given in the image of $V \otimes V^*$:

$$\begin{aligned} e_j \otimes \varphi_i &\mapsto \varphi_i(_ \cdot e_j) \in \mathcal{O}(G) \\ \forall X \in G, \quad \varphi_i(X \cdot e_j) &= \varphi_i\left(\sum_{k=1}^n X_{kj} e_k\right) = X_{ij} \end{aligned} \quad (2.4)$$

So, X_{ij} is also in the image of the embedding.

As a result, if consider the suitable tensor of the above representations, one can create arbitrary monomials in the image, and hence generate all polynomials in $\mathcal{O}(G)$. This is a way of explicitly proving that all generators of $\mathcal{O}(G)$ is in the image, hence the embedding is surjective.